Total No. of Printed Pages:01

SUBJECT CODE NO: - RNP-8018
FACULTY OF SCIENCE AND TECHNOLOGY
S.E (EEE / EE / EEP) (NEP) Sem - III
Examination May/June-2026
Electronics Devices & Circuits

[Time: 1:30 Hours] **[Max. Marks: 30]**

Please check whether you have got the right question paper.

N. B
1. Part A: Question No 1 is compulsory.
2. Attempt any FOUR questions from Part B: Q.2 to Q.7.
3. Use of non-programmable calculator is allowed

Part A

Q.1 Attempt ANY FIVE of the following questions **10**

a) In a PN junction diode, the width of depletion layer increases during:
   i) Forward bias
   ii) Reverse bias
   iii) Zero bias only
   iv) High temperature only

b) Define Q point & DC load line in BJT with suitable diagrams

c) Define a Frequency response & Bandwidth of an amplifier

d) Peak Inverse Voltage (PIV) of a diode is:
   i) Maximum forward voltage
   ii) Maximum reverse voltage it can withstand
   iii) Average output voltage
   iv) RMS voltage

e) Draw the symbols of Diode, BJT, FET & MOSFET

f) Sate the Bark Husain Criteria for Oscillations

g) Match the pairs:

Sr. No.	Parameters	Sr. No.	Options
1	BJT	a	Ripples remover
2	FET	b	AC to DC
3	Filter	c	Voltage controlled
4	Rectifier	d	Current controlled

Part B

Attempt ANY FOUR of the following questions

Q.2 Explain the Block diagram of DC regulated Power Supply **05**

Q.3 Define a Rectifier and explain all types of rectifiers with input & output waveforms. **05**

Q.4 Draw the circuit diagram of BJT based RC coupled amplifier **05**

Q.5 Define BJT Biasing? State the types of Biasing & explain any one **05**

Q.6 Differentiate between BJT, FET & MOSFET **05**

Q.7 Write a detailed note on High frequency oscillators **05**